

Time Series Fundamentals

for Stock Price Forecasting

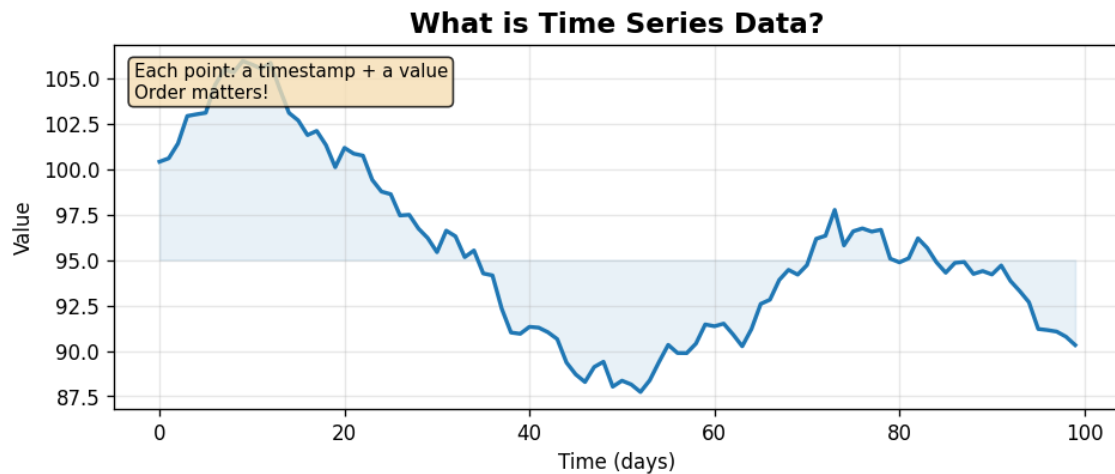
A beginner-friendly guide to understanding your notebook
Stock Price Forecasting with Apple (AAPL)

Written for Mursyid - May 2026

1. What is Time Series Data?

Time series data is simply data collected over time - where the ORDER matters. Unlike regular datasets (like a list of customers), you CANNOT shuffle time series data because each observation depends on what came before it.

Think of it like a movie: if you shuffle the frames, the story makes no sense. The same applies to stock prices - today's price depends on yesterday's price.



Examples of Time Series Data

- * Stock prices (AAPL) - what we're working with
- * Weather temperature readings
- * Website traffic over days
- * Your heart rate over time
- * Sales revenue monthly

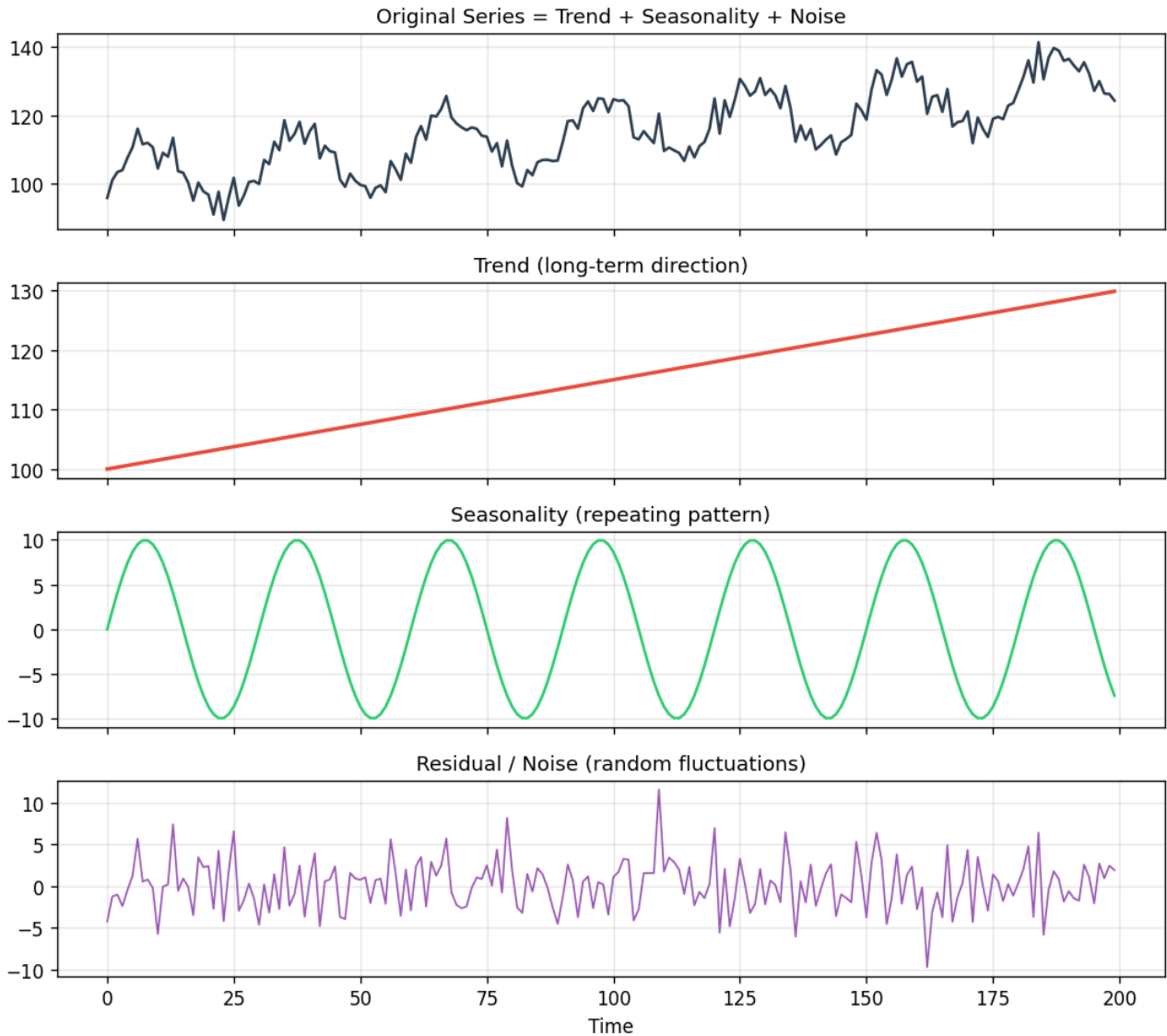
KEY INSIGHT for your notebook:

Stock prices are a specific type of time series called 'financial time series'.

They have: trends (long-term up/down), volatility (sudden spikes), and no fixed seasonality.

2. Components of a Time Series

Every time series can be broken down into three hidden components. Understanding these helps you decide what kind of model to use.



The Three Components

1. Trend

The long-term direction - is it going up, down, or sideways?

In your AAPL analysis: the trend was CLEARLY UP (Apple grew from ~\$80 to ~\$250 over 5 years).

Your notebook showed this with the 50-day and 200-day moving averages.

2. Seasonality

Regular, repeating patterns. Stock prices have WEAK seasonality (maybe more buying in January, or a 'Santa Claus rally' in December). But it's much weaker than, say, ice cream sales which peak every summer.

3. Residual / Noise

Random fluctuations you can't predict - news events, earnings surprises, tweets from CEOs.

This is why stock forecasting is HARD - there's always unpredictable noise.

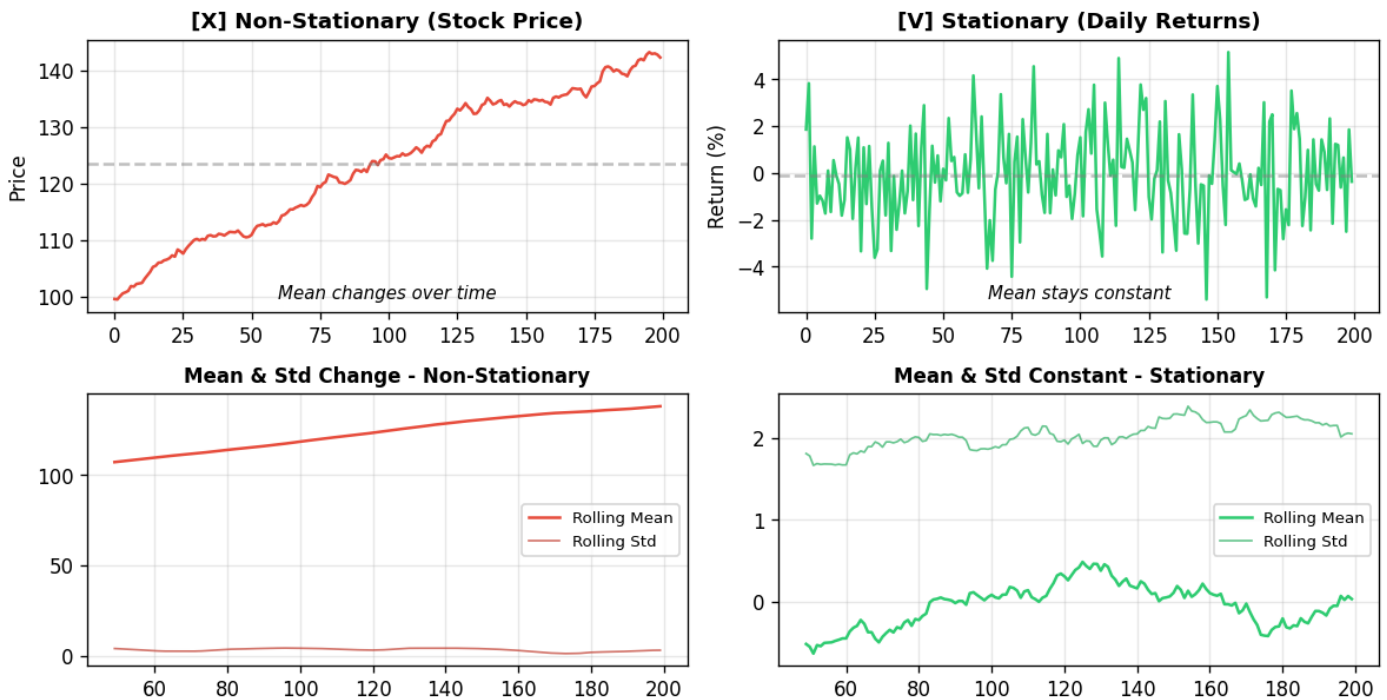
TIP: The EDA section in your notebook with ACF/PACF plots and the stationarity test all help identify these components. The moving averages (SMA_50, SMA_200) are your tool for isolating the trend.

3. Stationarity - The Most Important Concept

Stationarity is the single most important concept in time series analysis. If you understand this, you understand 80% of time series.

A stationary series has constant mean and constant variance over time. Stock prices are NOT stationary (they trend up). Stock RETURNS (% change day-to-day) ARE stationary.

Stationarity: The Most Important Concept in Time Series



Why Does Stationarity Matter?

- * Statistical models (ARIMA) REQUIRE stationarity - they'll give wrong answers otherwise
- * Tree-based models (Random Forest, XGBoost) DON'T need stationarity - one reason they work
- * Linear Regression partially needs it, but lag features often help mask non-stationarity

How to Check Stationarity (ADF Test)

Your notebook uses the Augmented Dickey-Fuller (ADF) test. Here's how to read it:

H0 (null hypothesis): Series is NON-stationary (has a unit root)

H1 (alternative): Series is stationary

p-value < 0.05 --> REJECT H0 --> Series IS stationary

p-value > 0.05 --> FAIL to reject H0 --> Series is NOT stationary

In your notebook results:

- . Close Price: p = 0.827 --> NON-STATIONARY (expected - prices trend up)
- . Daily Returns: p ~ 0.000 --> STATIONARY (returns bounce around zero)

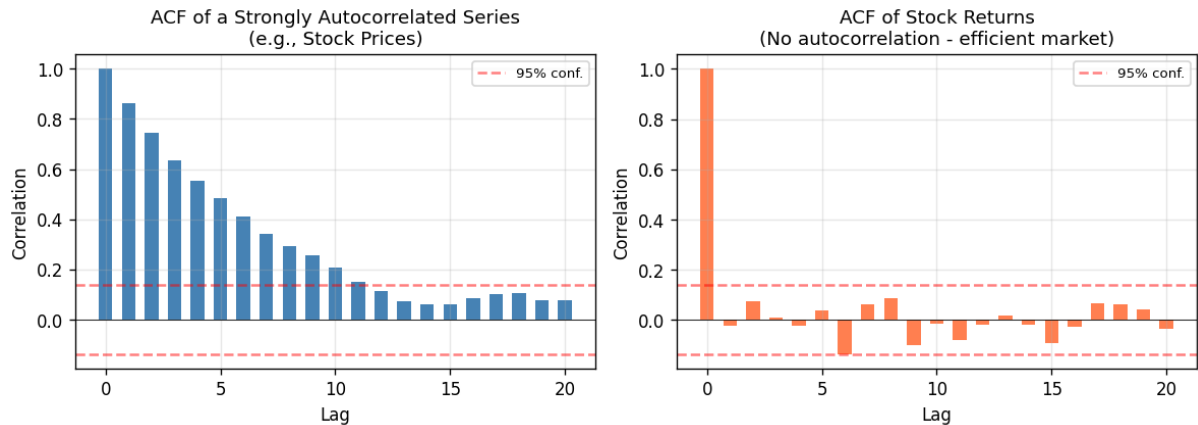
. 1st Difference: $p \sim 0.000$ --> STATIONARY (differencing removes the trend)

4. Autocorrelation - Does Past Predict Future?

Autocorrelation measures how a time series is correlated with ITS OWN PAST VALUES. In your notebook, this is shown through ACF and PACF plots.

If autocorrelation is high (like stock prices), then Lag_1 (yesterday's price) is a GREAT predictor of today's price. If autocorrelation is near zero (like stock returns), past returns don't help predict future returns - the market is 'efficient'.

Autocorrelation: Does Past Predict Future?



Key Terms

ACF (Autocorrelation Function)

Shows correlation at every lag. Slow decay --> non-stationary. Quick drop --> stationary.

PACF (Partial Autocorrelation)

Shows correlation at each lag AFTER removing the effect of shorter lags. Helps choose how many AR terms to use in ARIMA.

Lag

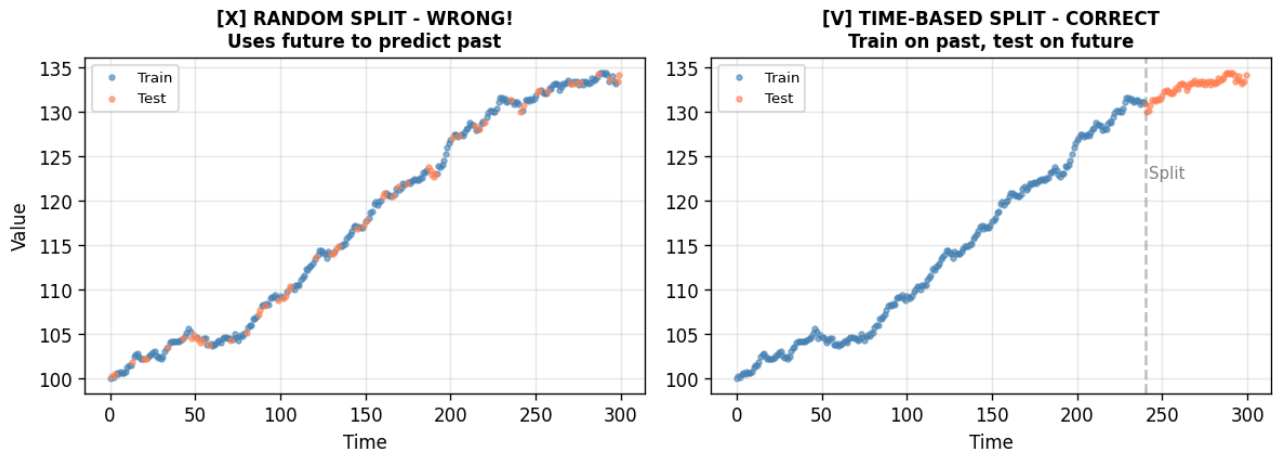
How far back we look. Lag 1 = yesterday, Lag 3 = 3 days ago, etc.

This is WHY your feature engineering creates Lag_1, Lag_2, Lag_3, Lag_5, etc.
Past prices ARE predictive of future prices - the ACF proves it!

5. The Golden Rule: No Look-Ahead Bias

In machine learning, we normally shuffle data and split randomly. IN TIME SERIES, YOU CANNOT DO THIS. This is the #1 beginner mistake.

Never Use Future Data to Predict the Past



In your notebook, we used an 80/20 TIME-BASED split:

Training: 2020-01-02 to 2023-12-29 (80% = 1,006 days)

Testing: 2024-01-02 to 2024-12-31 (20% = 252 days)

This means the model was trained ONLY on data before 2024, then tested on 2024 data - exactly like real forecasting.

Walk-Forward Cross-Validation

Your notebook also uses TimeSeriesSplit for a more robust check. Instead of one split, it creates 5 sequential train/test splits:

Fold 1: Train [0:204] -> Test [204:405]

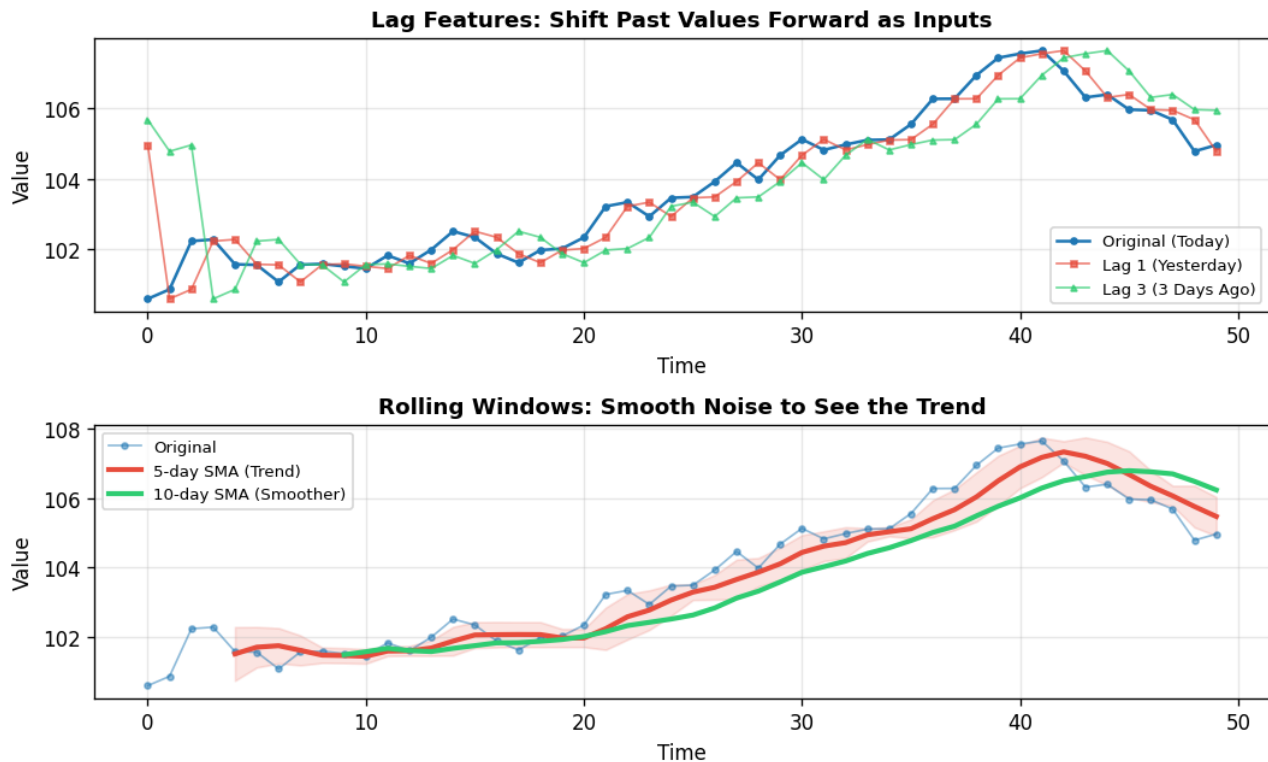
Fold 2: Train [0:405] -> Test [405:606]

...and so on, growing the training window each time.

This simulates how the model would have performed if deployed in real-time over the last 5 years.

6. Feature Engineering for Time Series

Features are the inputs our model learns from. In time series, we can't use today's Open/High/Low because we don't know them yet at prediction time. So we engineer features from the PAST.



The Feature Types in Your Notebook

Lag Features (Lag_1, Lag_3, Lag_5, etc.)

Shifted past values. Lag_1 = yesterday's close. These have the HIGHEST correlation with today's price - they're the most powerful features.

SMA (Simple Moving Average): SMA_5, SMA_10, SMA_20, SMA_50

Average of the last N days. Smooths out noise to show the trend. SMA_5 (1 week) is more responsive, SMA_50 (~2 months) shows the bigger picture.

RSI (Relative Strength Index): RSI_14

A momentum indicator from 0-100. Above 70 = 'overbought' (might fall). Below 30 = 'oversold' (might rise). Traders use this for buy/sell signals.

Rate of Change (ROC): ROC_5, ROC_10, ROC_20

Percentage change over N days. Positive = gaining momentum, Negative = losing momentum.

Price Ratios & Crossovers

Price / SMA_20 tells us if price is above or below the trend. SMA_5 / SMA_20 tells us if short-term trend is above or below medium-term trend.

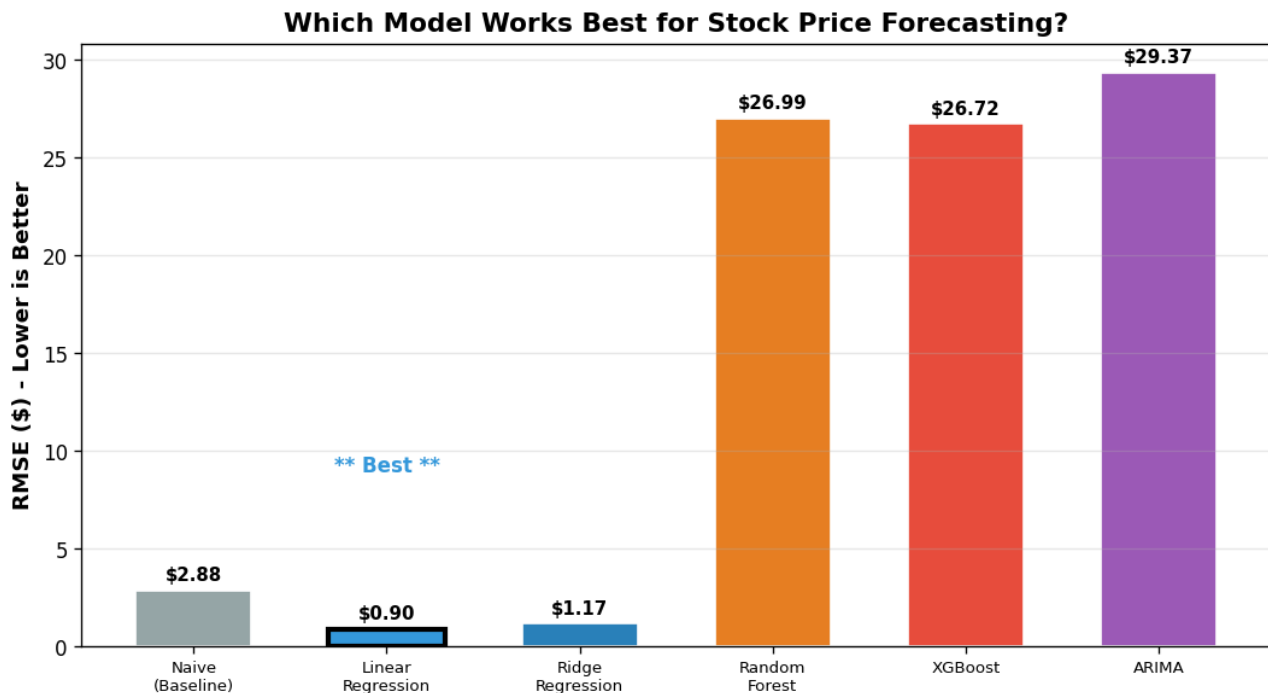
Volume Features

Trading volume helps confirm price moves. High volume + price up = strong move. Low volume + price up = weak move.

Calendar Features: Day_of_Week, Month

Stock markets sometimes show patterns by day (e.g., 'Monday effect') or month.

7. The Models - From Simple to Complex



Naive Baseline (The 'To Beat' Score)

The simplest forecast: 'tomorrow's price = today's price'. It's surprisingly good (RMSE = \$2.88) because stock prices are highly autocorrelated. Any real model must beat this.

Linear Regression (Your Best Model!)

RMSE = \$0.90 | MAPE = 0.32%

This won because lag features (Lag_1, Lag_3) and moving averages (SMA_5) are so strongly correlated with the target that a simple linear relationship captures most of the signal. It's fast, interpretable, and hard to beat when the right features exist.

Ridge Regression

RMSE = \$1.17 | MAPE = 0.41%

Same as Linear Regression but with L2 regularization - it penalizes very large coefficients to prevent overfitting. Slightly worse here because there's LITTLE risk of overfitting with only 29 features on 957 training samples.

Random Forest & XGBoost (Surprisingly Worse)

Random Forest: RMSE = \$26.99 | XGBoost: RMSE = \$26.72

These powerful models performed WORSE than linear models! Why?

1. Tree-based models struggle to EXTRAPOLATE beyond the range seen in training.
2. If the test set has prices higher than any in training, trees can't predict above that max.
3. Linear regression CAN extrapolate - it follows the trend line up.

This is a known limitation: trees work great for interpolation, poorly for extrapolation.

ARIMA

RMSE = \$29.37 | MAPE = 11.57%

The classic statistical model. It only uses the target's own past values - no features. ARIMA(5,1,5) means:

- . AR(5): Uses 5 past values
- . I(1): One round of differencing (to make it stationary)
- . MA(5): Uses 5 past forecast errors

It performed worst because it has NO feature engineering - just raw prices.

8. Glossary of Terms

ADF Test

Augmented Dickey-Fuller test. Statistical test for stationarity. $p < 0.05$ means stationary.

Autocorrelation

How a series correlates with its own past values.

Baseline

A simple model (like Naive) that shows the minimum acceptable performance.

Differencing

Subtracting each value from the previous one. Transforms non-stationary to stationary.

EWMA

Exponentially Weighted Moving Average. Like SMA but gives more weight to recent values.

Extrapolation

Predicting beyond the range of training data. Trees can't do this well.

Kurtosis

How 'fat' the tails are. Stock returns have kurtosis > 3 = more extreme events than normal.

Lag

A time-shifted version of a series. Lag 1 = yesterday's value.

Look-ahead bias

Using future information to predict the past. THE cardinal sin of time series.

MAE

Mean Absolute Error. Average $|\text{actual} - \text{predicted}|$. In dollars.

MAPE

Mean Absolute Percentage Error. Average error as a % of actual price.

Multicollinearity

When features are highly correlated with each other. Hurts linear models.

R-squared

Proportion of variance explained. 0.99 = model captures 99% of price movement.

Residuals

Actual - Predicted. Should be centered around zero (unbiased).

RMSE

Root Mean Squared Error. Like MAE but penalizes large errors more.

Rolling window

A fixed-size window that moves through time, computing stats at each step.

RSI

Relative Strength Index. 0-100 momentum indicator. Above 70 = overbought.

SMA

Simple Moving Average. Average of last N values. Smooths noise.

Stationarity

Constant mean and variance over time. Required for ARIMA.

Walk-Forward CV

Cross-validation that respects time order. Growing training window.

9. How This Connects to YOUR Notebook

Section-by-Section Mapping

Here's exactly what each part of your notebook does, mapped to concepts from this guide:

Data Fetching (yfinance)

This gives us real AAPL data. Time series needs real chronological data - not synthetic!

EDA - Price Trend

We plot the TREND component. The 50/200-day SMA isolates the long-term direction.

EDA - Distributions

We check for NORMALITY (kurtosis). Stock returns have 'fat tails' = more risk.

EDA - Stationarity (ADF)

THIS IS THE KEY TEST. It tells us prices are non-stationary but returns are stationary.

EDA - ACF/PACF

Proves that past prices (Lag features) are predictive. Guides ARIMA parameter selection.

Feature Engineering

Creates LAGS, SMA, RSI, ROC - all computed from PAST data only (no look-ahead!).

Naive Baseline

Sets the 'to beat' score. Any useful model must outperform 'predict tomorrow = today'.

Model Comparison

Shows WHY Linear Regression works best here (extrapolation ability) and why trees fail.

Walk-Forward CV

Simulates real-world deployment. High variance = model sensitive to market regimes.

30-Day Forecast

Iterative forecasting - each day's prediction becomes input for the next day.

FINAL ADVICE:

1. Always check stationarity first (ADF test) - it determines which models you can use.
2. NEVER shuffle time series data for train/test split.
3. Lag features are your most powerful tools - past prices DO predict future prices.
4. Linear models work surprisingly well for stock trends because they can extrapolate.
5. Tree-based models struggle with extrapolation - be aware of this limitation.
6. Combine price data with other sources (news sentiment, macro indicators) for better results.
7. Stock forecasting is HARD - even models with 99% R-squared can't predict sudden market moves.